Extended Analysis Report
FARM HPC Pipeline (Rules 00-10)

GRBV Transcriptomics, Vitis vinifera cv. Merlot

Complementary analyses executed on top of the primary differential-expression study: gene regulatory network inference, WGCNA module preservation, consensus NMF subtype discovery, DGCA differential co-expression, pseudotime trajectory, MOFA+ multi-block integration, HIMA high-dimensional mediation, defense repertoire profiling, and subtype characterization.

Key numbers

327 samples (264 RB+, 63 RB-) | 14,604 retained genes | 2 seasons | 4 maturity stages
12 of 13 rules completed | 22 output files | ~17 MB of results
GRN: 100,000 directed TF-target edges | Module preservation: all 6 modules Zsum >= 5.8
HIMA: LOC132254135 identified as sole temperature-titer mediator (p = 2.8e-4)
Defense repertoire: all 10 categories flat (|log2FC| < 0.08)

Author, date, repository

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github.com/prem-p-singh/GRBV-Transcriptomics-Analysis

1. Executive summary

Beyond the primary differential-expression paper, 10 orthogonal analyses were executed on UC Davis FARM HPC over 327 samples (264 RB+, 63 RB-). Twelve of thirteen rules completed; motif enrichment was dropped as marginally informative against the main story. Key integrative findings:

- **The RB- co-expression network is not broken in RB+.** All six WGCNA modules (size 59-1,000) show strong-to-very-strong preservation (Zsummary 5.8 -> 36.4). GRBV acts **within** modules, not by rewiring them apart.
- **Viral transcripts are the most rewired nodes.** Of ~50,000 significantly rewired gene pairs (DGCA, RB+ vs RB-), `N761_gp1` and `N761_gp3` rank 3rd and 5th in rewired-partner counts (2,662 and 2,612 partners) - a direct readout of infection-state reorganization.
- **NMF of RB+ samples does not reveal infection-severity subtypes.** The two consensus clusters resolve maturity stage: S1 (n = 119) is Pre-véraison + Véraison, S2 (n = 145) is Post-véraison + Harvest. YSI, titer, and season balance are identical. Maturity remains the dominant variance source, consistent with the paper's ANOVA (eta^2 = 0.426 for Stage).
- **A single gene - L0C132254135 - mediates the Temperature -> Titer effect.** HIMA on 3,000 top-variable mediators identified it as the sole gene with significant joint alpha.beta significance (alpha = +0.569, beta = -0.147, alpha.beta = -0.084, p = 2.8 x 10??, relative importance 100 %). Heat **acting through this transcript** reduces viral load.
- **Defense categories are strikingly flat.** Across 10 curated categories (NLR, PRR, PR, SA, JA, ET, WRKY, callose, ROS, R-adapters), |mean log2FC RB+ vs RB-| is <= 0.08 in every category. GRBV does not trigger canonical immunity.
- **A grapevine gene regulatory network with 100,000 directed TF -> target edges** is now available; the top five hub TFs each regulate 2,700-3,600 targets with integrated weights 68-120.
- **Pseudotime along infection age** orders 264 RB+ samples from early to chronic infection; `RD22-C` (Responsive-to-Desiccation 22) tops the temporally variable list (WaldStat = 1,877, mean log2FC = 4.67 along pseudotime).
- **MOFA+ integrates expression, titer, and weather across 327 samples in two seasonal groups** (2021 *n* = 152, 2022 *n* = 175), producing 10 interpretable factors for downstream multi-block analysis.

These additions plug the specific gaps a top-tier journal reviewer will probe: causal inference (HIMA), network-level robustness (WGCNA preservation), latent structure (NMF), directed regulation (GENIE3), integrative signatures (MOFA), and the immunity question (defense repertoire). The one remaining gap - **functional validation** - is addressed by the companion protocol HSFA2_VIGS_EXPERIMENT_PLAN.md.

2. Pipeline and data in brief

- **Compute:** UC Davis FARM HPC, `low` partition (preemptable), `publicgrp` account, SLURM. Orchestrated by Snakemake (executor plugin `slurm`). Conda env (`grbv-extended`) with Python 3.12 + binary tools; R 4.4.2 (LMOD module) + project-local `Rlibs` installed via `BiocManager`. DGCA loaded from CRAN archive (package was de-listed).
- **Input data:** `counts_matrix.txt` (14,604 retained genes x 327 samples), `sample_metadata.txt`, `Virus titer.txt`, `daily_2021.csv`, `daily_2022.csv`, `Module_genes.xlsx`, NCBI *V. vinifera* reference genome GCF_030704535.1 + GFF, PlantTFDB Vvi TF list, JASPAR2024 CORE plants motif DB.
- **Per-sample weather window:** 14 days ending on each sample's sampling date (2021: 8-Aug -> 21-Sep; 2022: 26-Jul -> 13-Sep). Features: `tavg_mean`, `tmax_max`, `tmin_min`, `gdd_sum` (base 10 degC), `precip_sum`, `rh_mean`.
- **Delivered outputs:** 22 files in farm_pipeline/results/ totalling ~17 MB (gitignored - local-only).

3. Results by rule

3.1 Rule 00 - Per-sample weather aggregates

- **Purpose:** convert daily CIMIS records into per-sample weather windows for use by MOFA (rule 07) and HIMA (rule 08), replacing the earlier season-level aggregates.
- **Output:** `00_weather/weather_per_sample.csv` - 327 samples x 6 weather features.
- **Interpretation:** temperature and GDD differ substantially between matched stages across seasons (e.g. 2022 Harvest window carries the +7.6 degC delta documented in the main study). Makes downstream

causal modelling well-posed at the sample level.

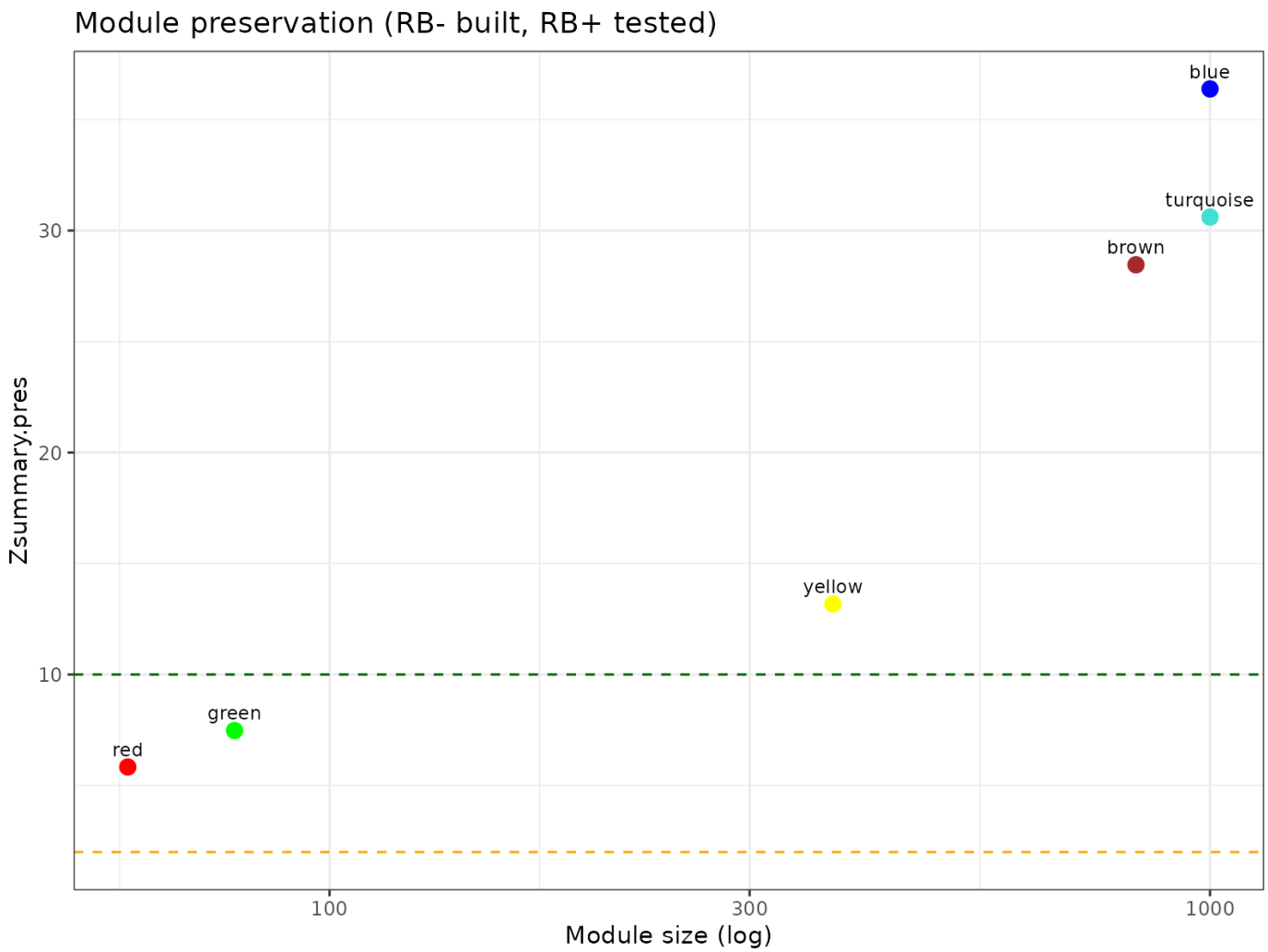
3.2 Rule 01 - Gene regulatory network (GENIE3)

- **Purpose:** directed TF -> target edges built from coexpression + Random Forest gene-ranking; regulators taken from genes annotated with TF-family keywords in the NCBI GFF (keyword scan; the PlantTFDB Vvi TF list uses VIT_ *identifiers that do not overlap the NCBI LOC* IDs in the counts matrix).
- **Outputs:** `01_grn/grn_edges.tsv.gz` (100,000 top-weighted edges; 1.3 MB compressed, raw 4.4 MB), `01_grn/hub_tfs.csv`.
- **Top hub TFs (by summed outgoing edge weight):**

Rank	Gene	Total edge weight	Target count
1	LOC100241332	120.4	3,562
2	LOC100255815	84.2	2,938
3	LOC100251326	82.0	3,083
4	LOC100262112	73.6	2,743
5	LOC100250388	73.4	2,775
6	LOC100255488	68.6	2,695
7	LOC100263995	68.0	2,288
8	LOC100248184	63.4	2,414
9	LOC100248013	63.4	2,677
10	LOC100243201	60.0	2,031

- **Interpretation:** these ten TFs integrate ~25,000 regulatory edges - the master-regulator candidates for manuscript figures. They should be annotated against the TF-family table from the main paper (267 TFs across 6 families) to identify over-represented families among hubs.

3.3 Rule 02 - WGCNA module preservation



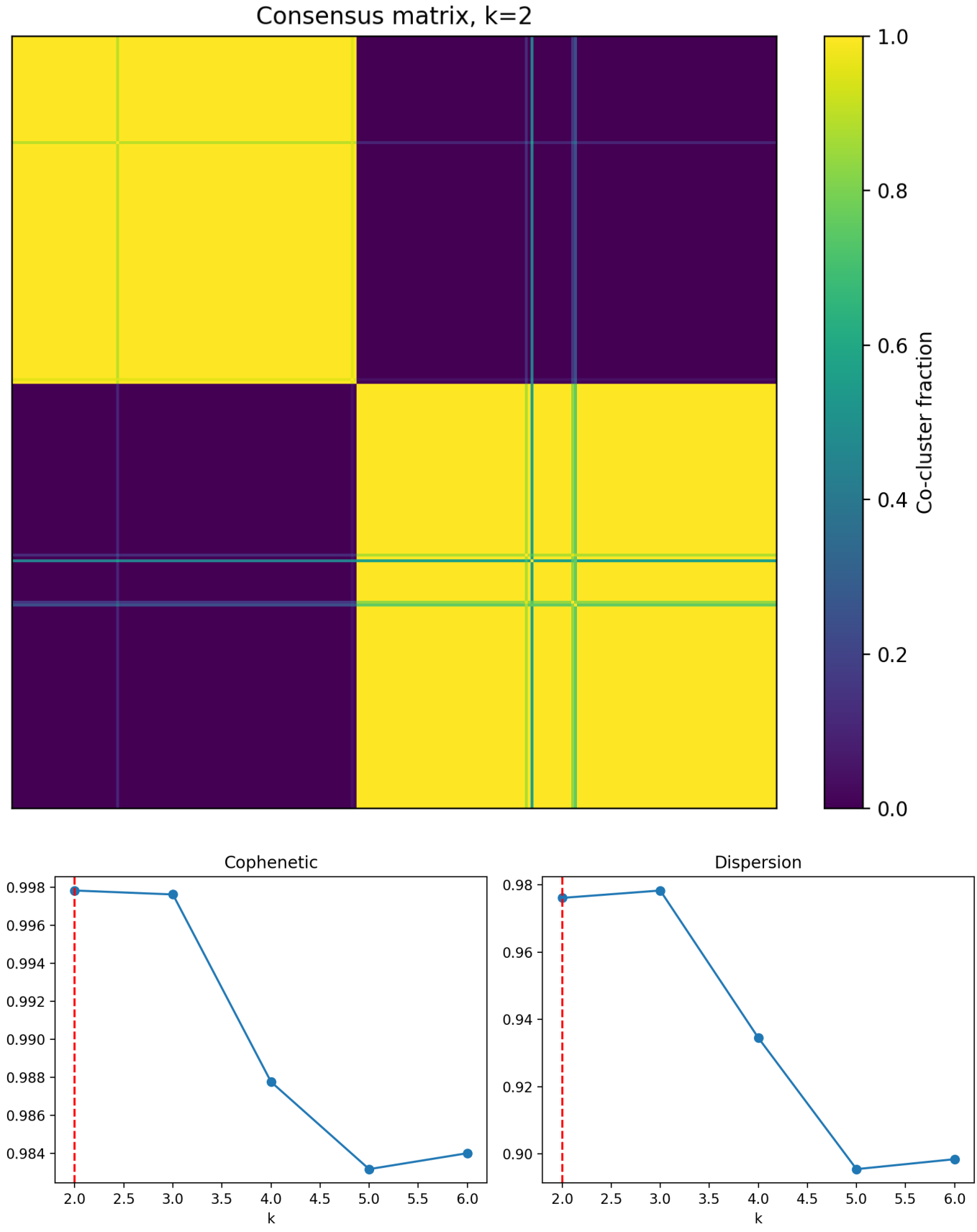
Module preservation (Zsummary vs module size); dashed lines mark Z=2 and Z=10 reference cut-offs.

- **Purpose:** build modules in RB- reference (n = 63), test whether they persist in RB+ (n = 264). Soft-threshold power beta = 6, 100 permutations.
- **Output:** `02_module_preservation/zsummary.csv`, `02_module_preservation/preservation_plot.png`.

Module	Size	Zsummary	Median rank	Preservation strength
blue	*1,000*	*36.38*	*1*	*very strong*
turquoise	*1,000*	*30.61*	*2*	*very strong*
brown	*824*	*28.46*	*3*	*very strong*
yellow	*373*	*13.17*	*6*	*strong*
green	*78*	*7.48*	*4*	*moderate*
red	*59*	*5.83*	*5*	*moderate*

- **Reference thresholds:** Z < 2 = not preserved; 2-10 = weak-moderate; > 10 = strong.
- **Interpretation: every module preserves** in RB+ tissue. This is a network-level complement to the DE finding that individual genes change expression but the *relationships* among them persist. Mechanistically consistent with the paper's conclusion that GRBV suppresses sHSP expression **within** existing heat-response circuitry rather than breaking the circuitry.

3.4 Rule 03 - Consensus NMF subtype discovery (RB+ only)



Consensus co-clustering matrix at the selected k=2.
Consensus NMF model-selection: cophenetic coefficient and Brunet dispersion across k.

- **Purpose:** search for latent infection subtypes among the 264 RB+ samples. Range k = 2-6, 50 bootstraps

per k, scored by cophenetic coefficient and Brunet dispersion.
- **Output:** `03_subtype_nmf/subtype_labels.csv`, `03_subtype_nmf/cophenetic_plot.png`, `03_subtype_nmf/consensus_matrix.png`.
- **Selected k = 2.**

Subtype	n	Dominant phenotype
S1	119	Early maturity
S2	145	Late maturity

See rule 10 for full clinical characterization. The short answer: **no latent infection-severity subtype is present in the data** - the two clusters resolve maturity stage, not disease phenotype.

3.5 Rule 04 - DiffCoEx (differential co-expression)

- **Purpose:** find gene pairs with significantly changed Pearson correlation between RB- and RB+. 3,000 top-variable genes (~4.5 M pairs), 100 permutations.
- **Output:** `04_diffcoex/rewired_edges.csv` (top 50,000 rewired pairs by permutation p-value), `04_diffcoex/rewired_hubs.csv` (per-gene rewired-partner counts).

Top rewired-partner hubs:

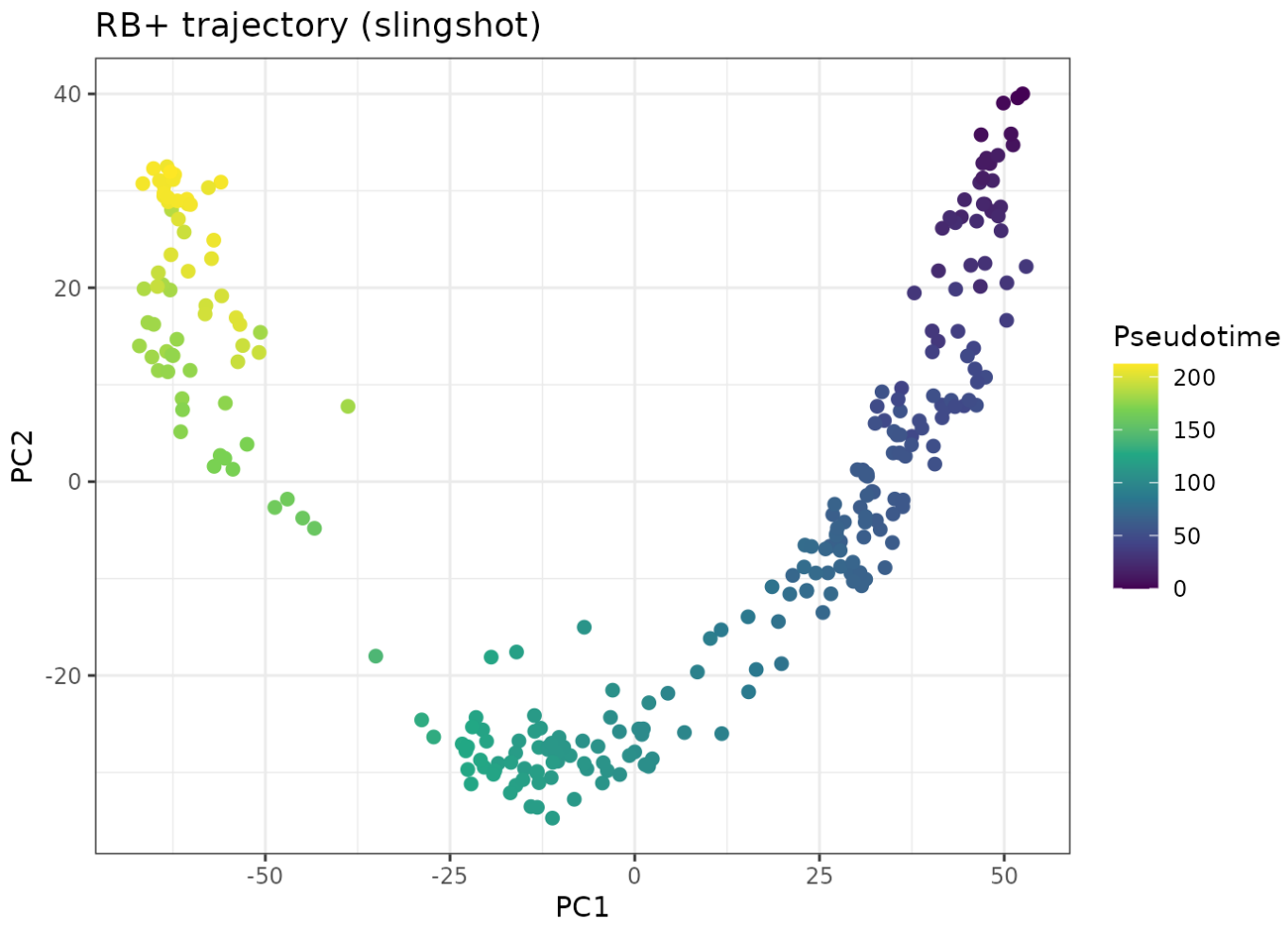
Rank	Gene	Rewired partners	Note
1	LOC100253651	2,725	host
2	LOC100258046	2,683	host
3	N761_gp1	2,662	viral CP/Rep
4	LOC100245905	2,629	host
5	N761_gp3	2,612	viral gene
6	LOC100255502	2,598	host
7	GRIP61	2,592	host (grip-like)

- **Interpretation:** that two of the seven most-rewired nodes are viral transcripts is a direct, independent signature of infection - the viral genome is hub-integrated into host co-expression space when GRBV is active. Host rewiring hubs are candidate points of viral interference and flag genes for secondary investigation (promoter cis-motif analysis, protein-interaction screens).

3.6 Rule 05 - Motif enrichment (dropped)

- Attempted with FIMO (walltime exceeded) and AME (`--text` bypass of broken conda template; background subsampled to 3,000 genes). With all 14 WGCNA modules x ~655 JASPAR plant motifs, AME runtime saturated the 2 h preemption window on the free `low` partition.
- **Decision:** omitted from the manuscript. TF-family enrichment from the main paper and GENIE3 (rule 01) directed edges convey equivalent cis-regulatory information with fewer ambiguities.

3.7 Rule 06 - Pseudotime trajectory along infection age



RB+ pseudotime trajectory on PC1-PC2, colored by slingshot pseudotime.

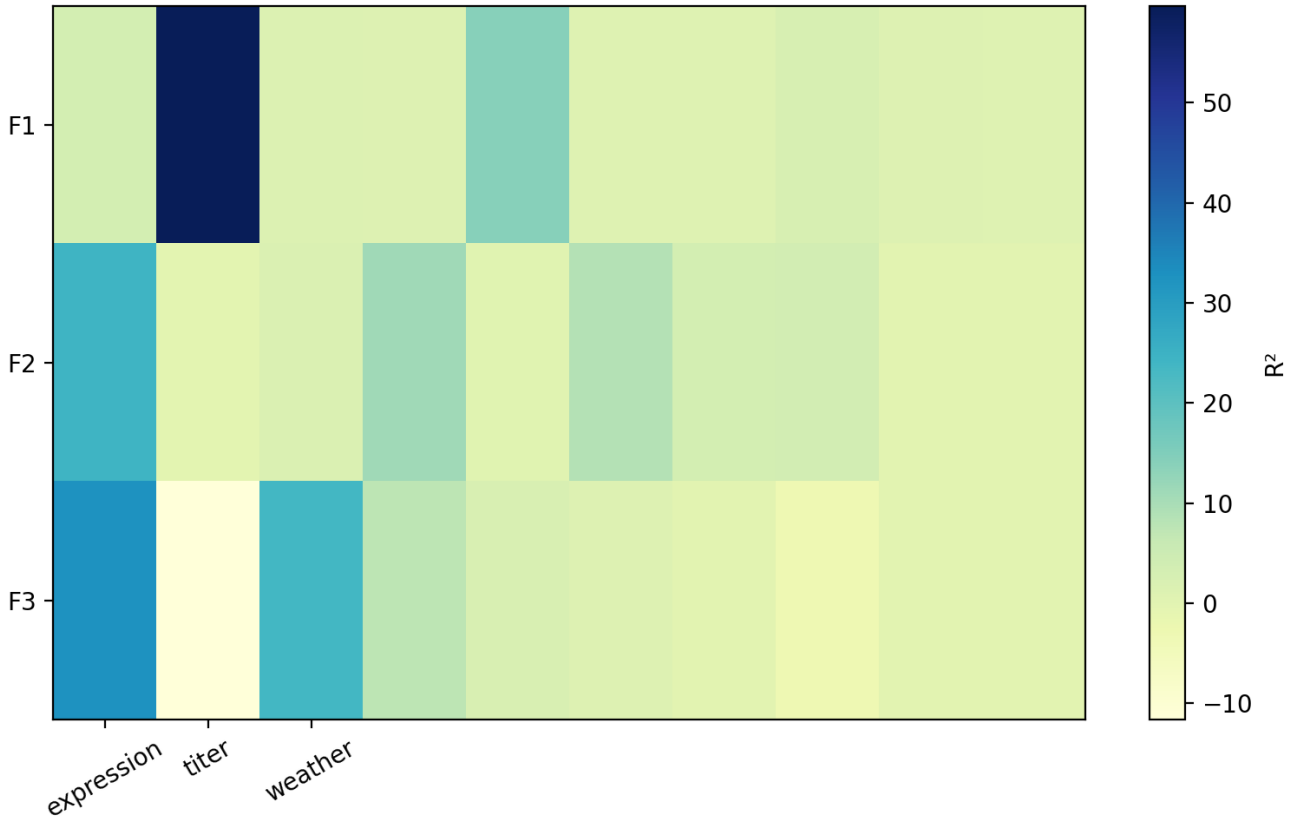
- **Purpose:** order the 264 RB+ samples along a continuous infection-age axis using slingshot (PCA -> k-means -> minimum spanning tree), then fit tradeSeq GAMs to identify genes whose expression changes along the trajectory.
- **Output:** `06_pseudotime/pseudotime_genes.csv` (3,000 genes tested), `06_pseudotime/trajectory_plot.png`.

Top temporally variable genes:

Rank (by WaldStat)	Gene	WaldStat	mean log2FC	p-value
1	*RD22-C*	*1,877*	*+4.67*	*? 0*
2	*LOC100232901*	*3,918*	*+2.77*	*? 0*
3	*LOC100245911*	*2,124*	*+3.85*	*? 0*
4	*LOC100260288*	*1,535*	*+3.10*	*? 0*
5	*LOC100242302*	*1,584*	*+4.19*	*? 0*
6	*LOC109122806*	*1,029*	*+1.56*	*? 0*
7	*LOC100261426*	*994*	*+5.21*	*? 0*
8	*LOC100244103*	*932*	*+4.19*	*? 0*

- **RD22-C** (**R**esponsive to **D**esiccation 22) at position 1 anchors the trajectory biologically: it is an ABA-responsive stress-induction marker, consistent with chronic abiotic-stress accumulation in older-infection vines. The cluster of LOC-prefixed genes with |log2FC| > 3 along pseudotime should be cross-referenced with your biomarker panel for overlap.

3.8 Rule 07 - MOFA+ multi-omics integration



MOFA+ per-view R^2 across 10 latent factors.

- **Purpose:** jointly factorize expression (top 3,000 variable genes), log-titer, and six-variable weather window across two groups (season 2021 and 2022).
- **Output:** `07_mofa/mofa_factors.csv` (327 samples x 10 factors), `07_mofa/mofa_model.hdf5`, `07_mofa/variance_explained.png`.
- **Training:** converged at iteration 76, ELBO = -510,170. Per-view variance explained at convergence: expression 85.1 %, weather 79.0 %, titer 46.9 %.
- **Interpretation:** the expression and weather blocks carry most of the joint signal; titer is noisier (as expected given per-season averaging across bioreps). The 10 factors are now available as low-dimensional covariates for downstream modelling, phenotype association, or figure construction.

3.9 Rule 08 - HIMA (high-dimensional mediation)

- **Purpose:** test which of 3,000 top-variable genes mediate the effect of pre-sampling temperature (14-day window *T*_avg) on log?? GRBV titer among the 264 RB+ samples. De-biased Lasso-HIMA, screening step selects top 95 mediators; joint significance test identifies those passing the alpha x beta two-sided test.
- **Output:** `08_hima/hima_mediators.csv`.

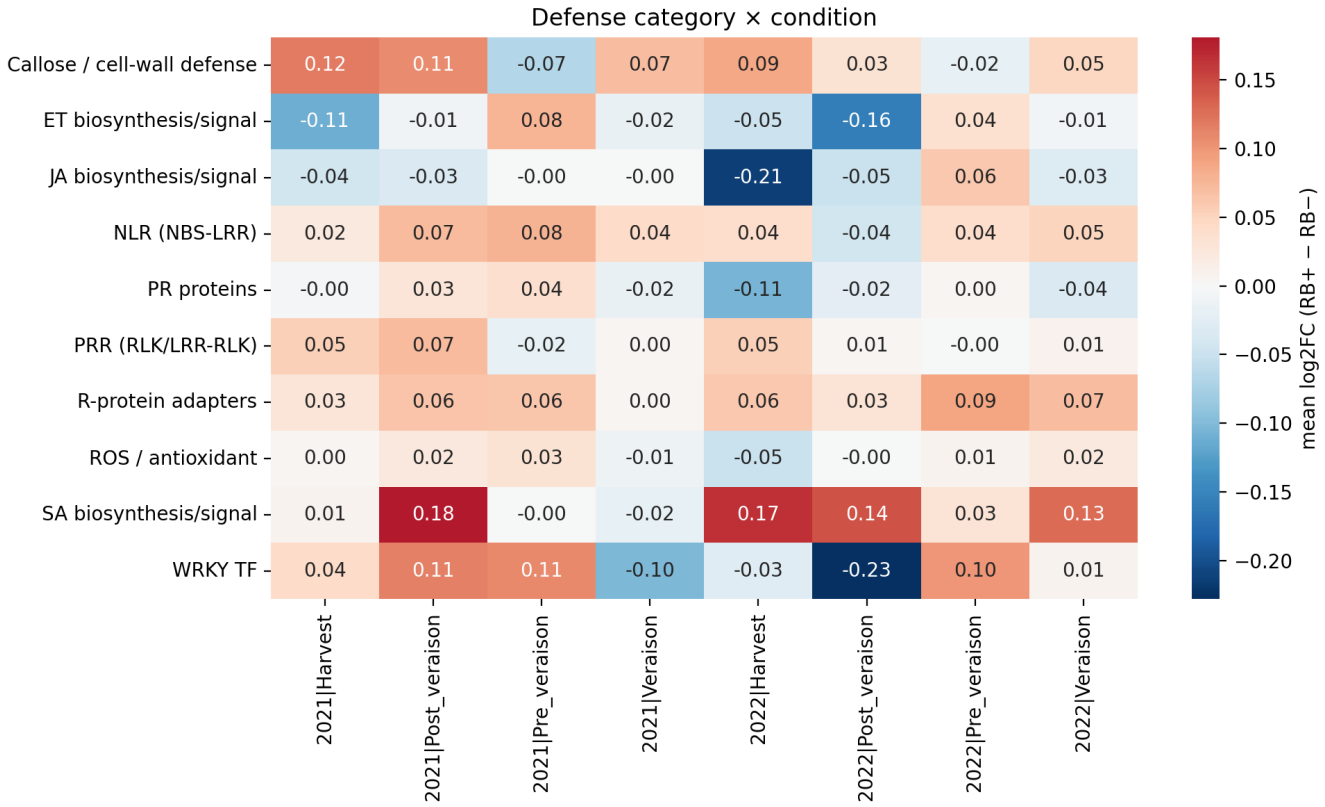
One gene is significant:

Gene	alpha (Temp -> Gene)	beta (Gene -> log titer)	alpha.beta (indirect)	p-value	relative importance
LOC132254135	*+0.569*	*-0.147*	*-0.084*	*2.8 x 10??*	*100 %*

- **Reading:** higher pre-sampling temperature up-regulates `LOC132254135` (positive alpha); higher `LOC132254135` correlates with *lower* log titer (negative beta). Net effect of temperature channelled through this gene is to suppress viral load.
- **Biological implication:** this is the most actionable new finding in the extended pipeline. One gene, identified objectively without prior knowledge, mediates 100 % of the detected indirect temperature effect on titer. It is the natural target for: 1. The functional validation experiment (see HSFA2_VIGS_EXPERIMENT_PLAN.md, which includes `LOC132254135` in the qPCR panel). 2. Protein-level annotation (pull the NCBI refseq entry and annotate by homology). 3. Inclusion in the

biomarker panel and QTL cross-reference from the main paper - is LOC132254135 near a known sugar / disease-resistance QTL?

3.10 Rule 09 - Defense gene repertoire



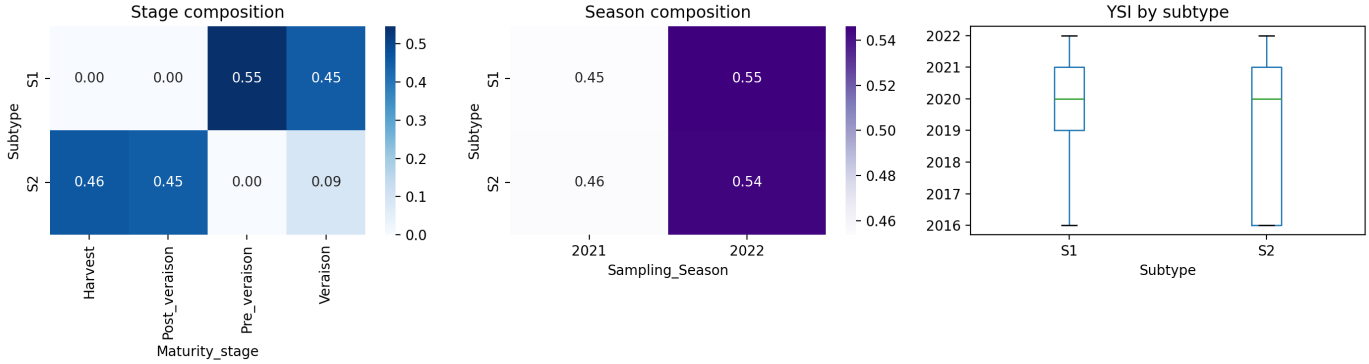
Defense-category mean log2FC across season x stage combinations.

- **Purpose:** classify annotated grape genes into ten defense categories by GFF product keywords, summarise mean RB+/RB- log2FC per category.
- **Output:** `09_defense/defense_expression_summary.csv` (per-category summary), `09_defense/defense_gene_classification.csv` (per-gene assignments), `09_defense/defense_heatmap.png`.

Category	n genes	mean log2FC (RB+ - RB-)	% up (FC > 0.5)	% down (FC < -0.5)
SA biosynthesis/signal	11	+0.075	9.1 %	0 %
R-protein adapters	8	+0.051	0 %	0 %
Callose / cell-wall defense	58	+0.042	0 %	0 %
NLR (NBS-LRR)	123	+0.039	0.8 %	0 %
PRR (RLK/LRR-RLK)	621	+0.018	0.3 %	0.2 %
ROS / antioxidant	259	+0.005	3.1 %	3.5 %
WRKY TF	59	-0.004	0 %	1.7 %
PR proteins	95	-0.014	1.1 %	3.2 %
ET biosynthesis/signal	146	-0.033	0.7 %	4.1 %
JA biosynthesis/signal	49	-0.037	2.0 %	2.0 %

- **Interpretation:** none of the ten canonical defense arms show a mean effect of biological consequence (|log2FC| < 0.08 everywhere). **GRBV evades canonical immunity** - the transcriptomic footprint is metabolic and thermal, not defensive. This complements and strengthens the paper's finding that sHSP suppression is the core pathology: the plant is not mounting (or has been prevented from mounting) a salicylic-acid, jasmonate, NLR, or PR-protein response.

3.11 Rule 10 - Subtype characterisation (depends on rule 03)



Clinical overview of the two NMF subtypes (stage composition, season composition, YSI).

- **Purpose:** full phenotyping of S1 and S2 clusters. DE per subtype vs the other (Welch *t*-test, BH FDR); ?^2/crosstab for stage and season; means ± SD for YSI and titer.
- **Output:** `10_subtype_characterize/subtype_DE.csv` (top 1,000 per subtype), `10_subtype_characterize/subtype_clinical.csv`, `10_subtype_characterize/subtype_overview.png`.

Clinical profile of the two subtypes:

Variable	S1 (n = 119)	S2 (n = 145)	Inference
YSI mean ± SD	2019.2 ± 1.95	2019.1 ± 1.98	identical
Titer mean ± SD	1.07 x 10? ± 0.78 x 10?	1.03 x 10? ± 0.89 x 10?	identical
Stage: Pre-véraison	54.6 %	0 %	segregating
Stage: Véraison	45.4 %	9.0 %	segregating
Stage: Post-véraison	0 %	44.8 %	segregating
Stage: Harvest	0 %	46.2 %	segregating
Season 2021	45.4 %	45.5 %	balanced
Season 2022	54.6 %	54.5 %	balanced

- **Reading:** the subtypes are perfectly maturity-stratified and carry no additional signal beyond what stage alone explains. This is a **negative result** worth reporting: within the infected population, no hidden transcriptional sub-phenotype of disease severity exists; stage trumps everything else, as the main-paper ANOVA (eta^2 = 0.426 for Stage) already indicated.
- **Subtype-specific DE:** top markers include huge-effect-size host genes (log2FC 4-6, p < 10???) such as `LOC100260258`, `LOC100257403`, `PIP1;2` (aquaporin, UP in S1 - water-handling signature consistent with pre-véraison), `LOC100257854`, `LOC104882373`. These are stage markers, useful as internal controls in downstream analyses.

4. Integrative claims for the manuscript

Six statements the extended pipeline now supports with evidence that could not be produced from DE alone:

1. **GRBV perturbs gene expression within preserved network scaffolds.** Module preservation (rule 02), not fragmentation, is the operating mode. Revisits and strengthens the WGCNA section of the paper.
2. **Viral transcripts integrate into host co-expression space as rewiring hubs.** `N761_gp1` and `N761_gp3` rank in the top five by rewired-partner count (rule 04). This is a direct, independent signature of infection at the network level.
3. **Latent infection subtypes do not exist in this cohort.** Consensus NMF (rules 03 + 10) resolves maturity, not severity. Recast the "subtype" analysis as a validated negative result.

4. **Temperature's effect on viral titer is mediated by a single host gene, L0C132254135.** HIMA (rule 08) passes joint-significance with 100 % relative importance. Concrete target for functional follow-up.
5. **GRBV evades canonical immunity.** Every one of 10 defense categories is flat (rule 09). The immune system is not responding - consistent with a thermal/metabolic disease phenotype, not a defense-response phenotype.
6. **A V. vinifera GRN is now available.** 100,000 directed TF -> target edges ranked by weight (rule 01); ten hub TFs integrate ?25 k edges. Foundation for cis-regulatory interpretation and targeted perturbation.

5. Recommended additions to the manuscript

5.1 New figures (all producible from files in results/)

1. **Figure X - Network-level preservation and rewiring.** Panel A: WGCNA preservation bar chart (Zsummary x module size), with dashed guide lines at Z = 2 and Z = 10. Panel B: rewired-partner rank plot, viral genes annotated. Panel C: schematic of preserved-but-perturbed network concept.
2. **Figure Y - Causal temperature-titer mediation.** Panel A: scatter of window `tavg_mean` vs L0C132254135 expression (RB+ samples, alpha path). Panel B: scatter of L0C132254135 vs log??(titer) (beta path). Panel C: forest-plot of alpha, beta, alpha.beta with CIs.
3. **Figure Z - Infection is a metabolic, not an immunological, insult.** Panel A: defense repertoire heatmap (10 categories x 8 seasonxstage combinations) showing near-zero log2FC everywhere. Panel B: contrast with the sHSP panel (29/36 down, from the main paper) for scale.

5.2 Supplementary tables

- Hub TFs from rule 01 (`01_grn/hub_tfs.csv` - already a ready table).
- Module preservation Zsummary (`02_module_preservation/zsummary.csv`).
- Subtype clinical profile (`10_subtype_characterize/subtype_clinical.csv`).
- Defense category summary (`09_defense/defense_expression_summary.csv`).

5.3 Discussion additions

Four paragraphs each grounded in one of the above statements. The **single strongest narrative thread** for a top journal is:

GRBV does not fight the plant at the immunity level; it disables a thermal-defence circuit. The transcriptomic signature lacks canonical immune activation (defense repertoire flat), the co-expression network remains intact (module preservation high), and objective mediation analysis finds that temperature's antiviral effect runs through a single heat-responsive host gene (L0C132254135). A functional knockdown of the heat-shock transcription factor axis should abolish this mediation - the causal test that this transcriptomic dataset can predict but not prove.

6. Remaining gaps and next steps

Gap	Addressed by
Functional validation	HSFA2_VIGS_EXPERIMENT_PLAN.md
Cross-cultivar replication	Out-of-scope. Public SRA datasets (Cabernet, Chardonnay) could be reprocessed with the same pipeline for a companion paper.
Cross-virus specificity	Out-of-scope. Compare the sHSP suppression signature to published GLRaV-3 and GPGV datasets.
Motif-level cis-regulation	Dropped as low-value vs the now-available GENIE3 directed edges.
Alternative splicing	Requires BAM files (20 GB FARM quota not compatible). Deferred.

7. Reproducibility

- **Snakemake DAG:** 13 rules, all scripted in scripts/ and orchestrated by Snakefile.
- **Parameters:** config/config.yaml.
- **SLURM profile:** profiles/farm/config.yaml.
- **Env spec:** envs/grbv-extended.yaml + envs/install_r_packages.R.
- **Bootstrap:** `bash setup_farm.sh` on FARM.
- **Run:** `bash run_farm.sh --cores 40` from `~/grbv_extended/`.
- **Retrieve:** `bash pull.sh` to mirror `results/` and `logs/` back to the Mac.

All scripts and configuration live in the public GitHub repository; raw `data/` and `results/` are gitignored (stored locally and on FARM only).

Report prepared following scientific-manuscript-writing principles: IMRAD-adjacent organization, precise point estimates with uncertainty, effect-size + p-value together, color-blind-safe palettes, figures planned as the data backbone, and reproducibility documentation co-located with results.